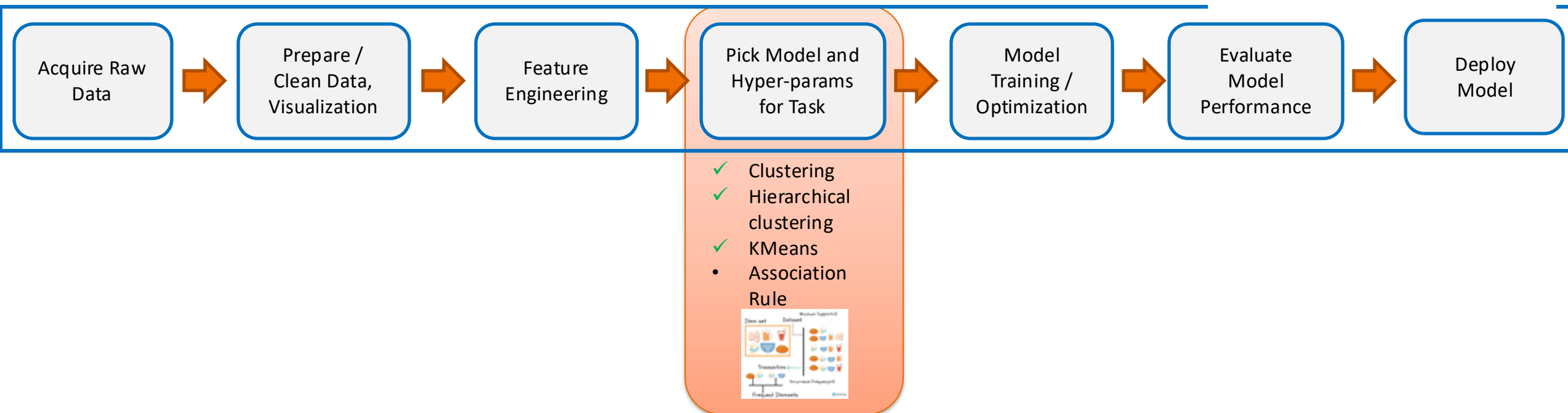




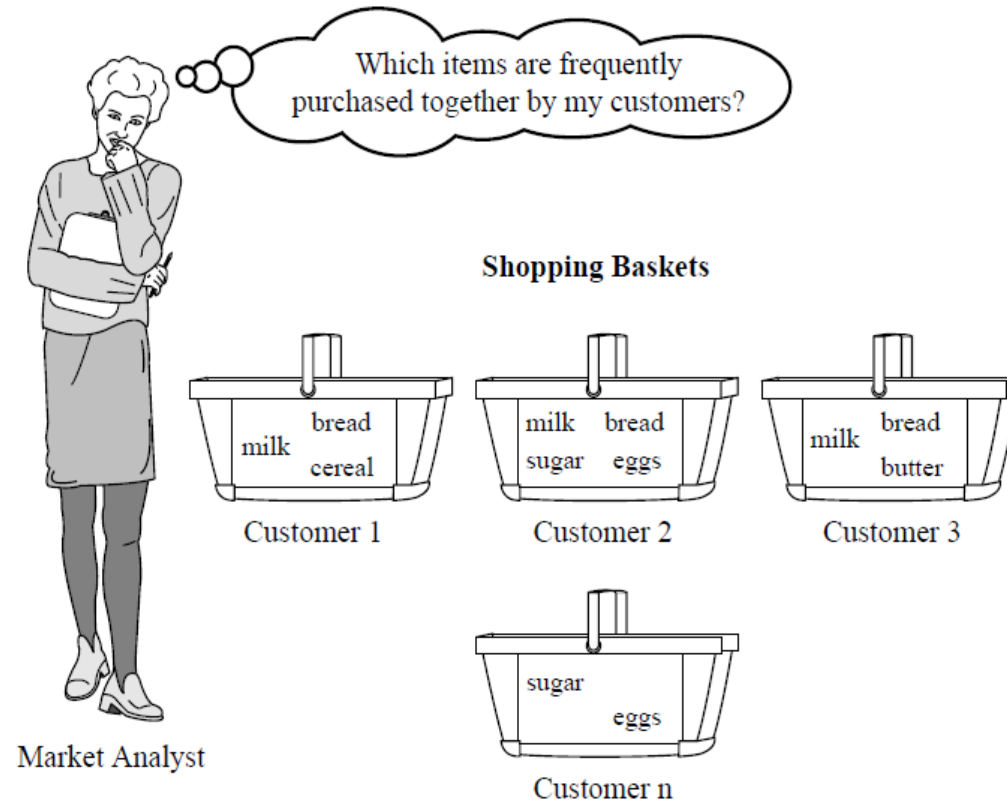
Identifying Co-occurring Patterns

Association Rules

What Is Frequent Pattern Analysis?

- **Frequent pattern:** a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of **frequent itemsets and association rule mining**
- Motivation: Finding inherent regularities in data
 - What products were often purchased together? — bread and butters?!
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
- Applications
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

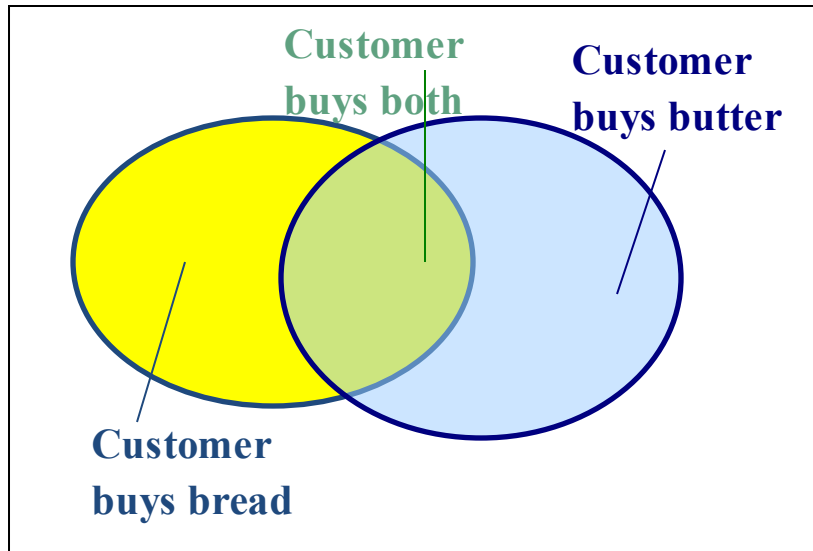
Market Basket Analysis



computer $\Rightarrow$ antivirus software [support = 2%, confidence = 60%]

Basic Concepts: Frequent Patterns

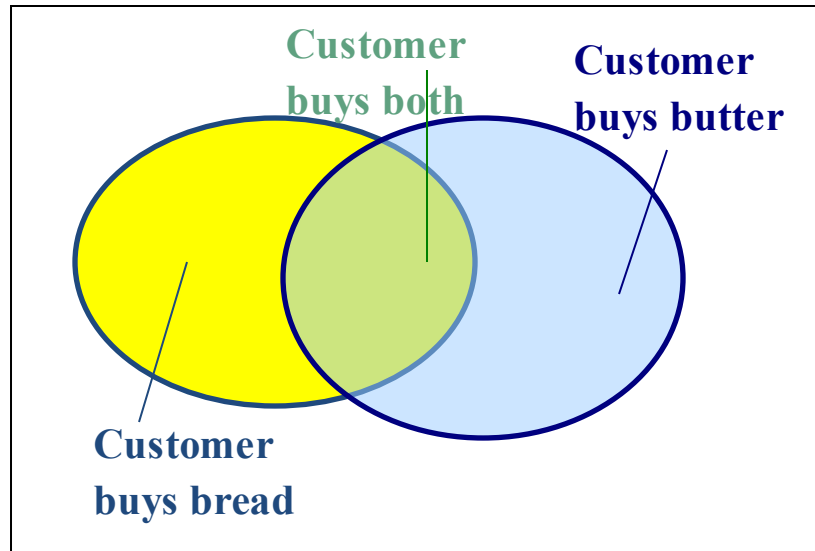
Tid	Items bought
10	bread, Nuts, butter
20	bread, Coffee, butter
30	bread, butter, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, butter, Eggs, Milk



- **itemset**: A set of one or more items
- **k-itemset** $X = \{x_1, \dots, x_k\}$
- **(absolute) support, or, support count** of X : Frequency or occurrence of an itemset X
- **(relative) support**, s , is the fraction of transactions that contains X (i.e., the **probability** that a transaction contains X)
- An itemset X is **frequent** if X 's support is no less than a minsup threshold

Basic Concepts: Association Rules

Tid	Items bought
10	bread, Nuts, butter
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- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - support, s , probability that a transaction contains $X \cup Y$
 - confidence, c , conditional probability that a transaction having X also contains Y . $P(Y|X)$

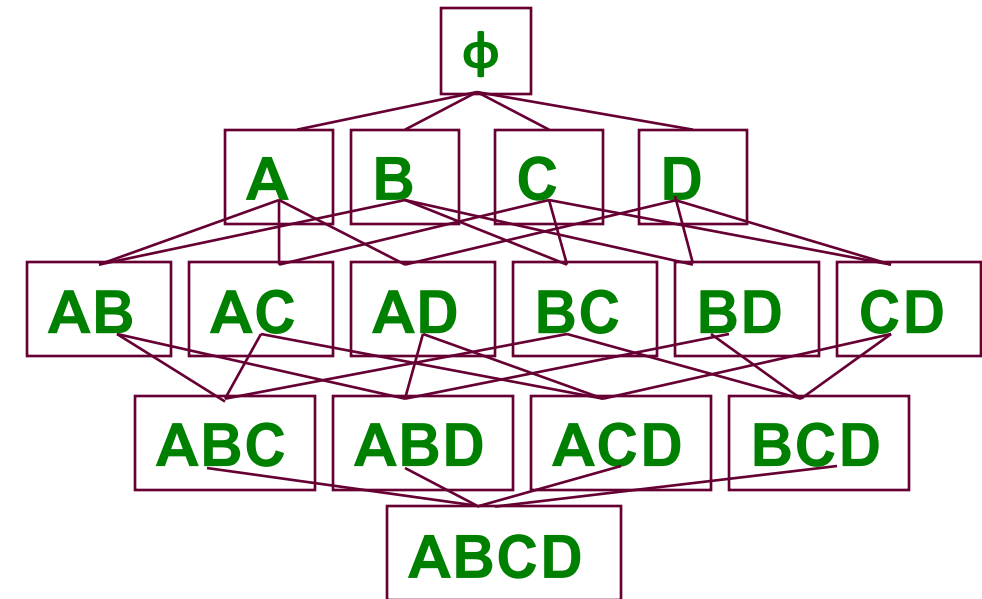
Let minsup = 50%, minconf = 50%

Freq. Pat.: bread: 3, Nuts: 3, butter: 4, Eggs: 3, {bread, butter}: 3

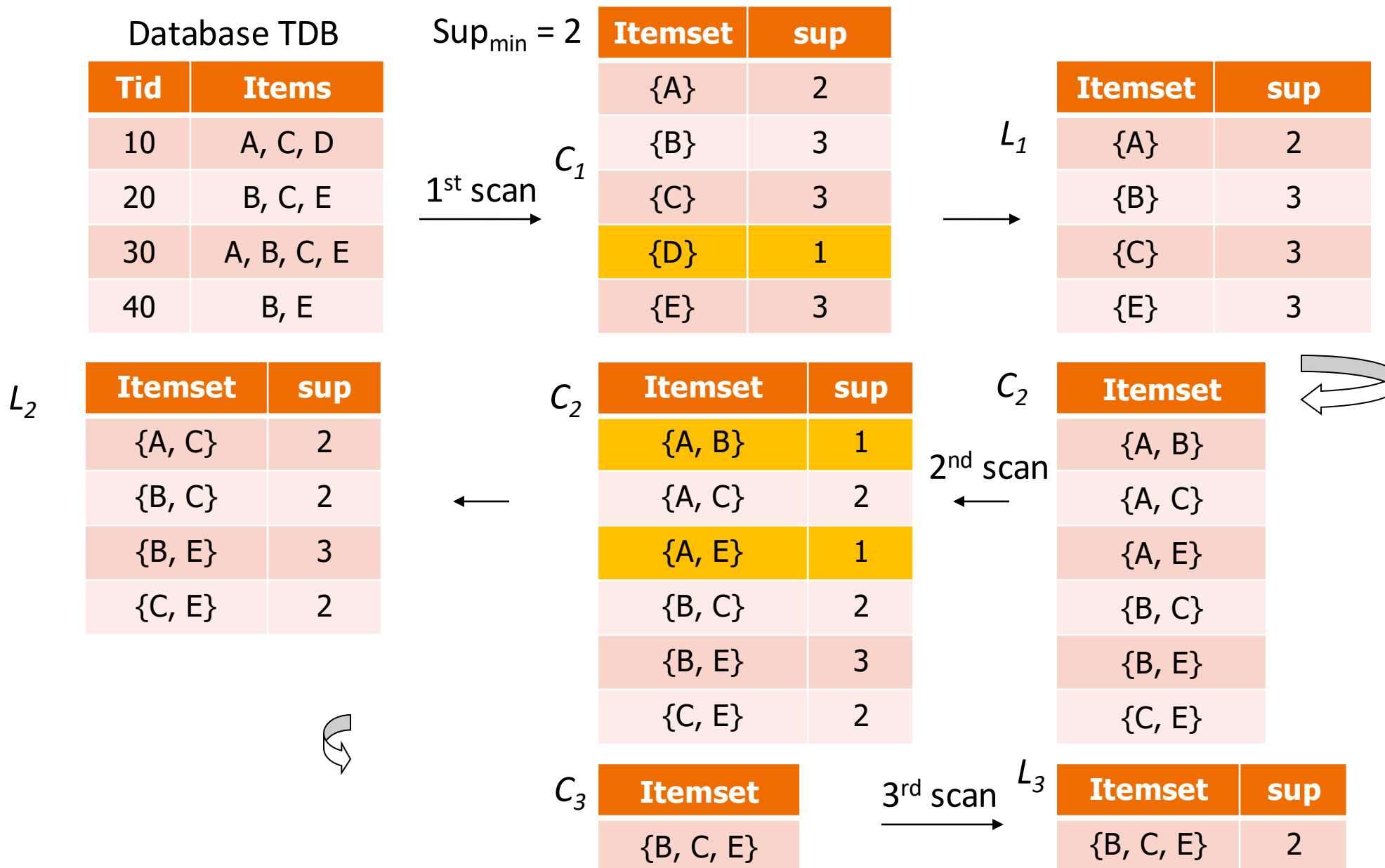
- Association rules:**
 - bread $\rightarrow$ butter (60%, 100%)
 - butter $\rightarrow$ bread (60%, 75%)

Difficulty

- Extremely computationally expensive
- Naïve solution
 - exponential time and memory w.r.t. $|I|$
 - linear time w.r.t. $|D|$
- Typically, $|I|$ is in thousands, $|D|$ is in billions...



The Apriori Algorithm—An Example

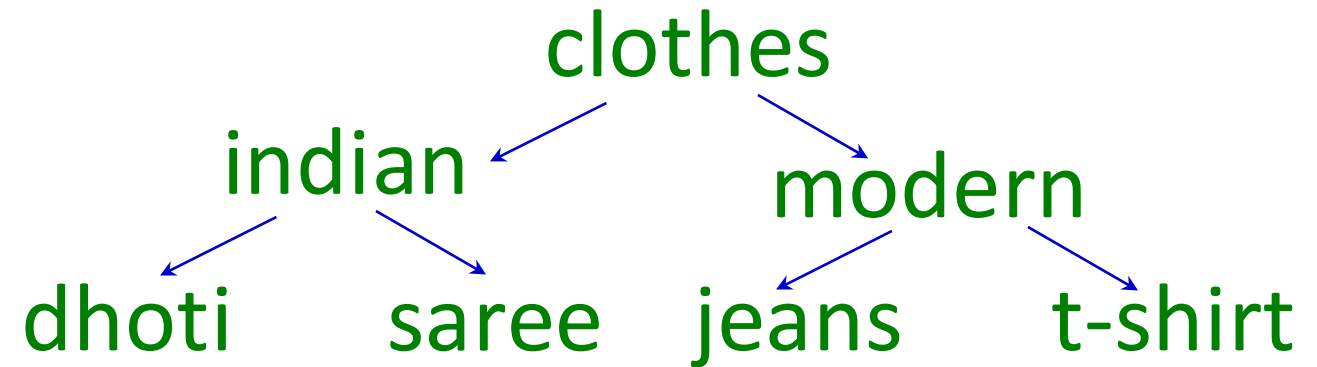


Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is any itemset which is infrequent, its superset should not be generated/tested! (Agrawal & Srikant @VLDB'94, Mannila, et al. @ KDD' 94)
- Method:
 - Initially, scan DB once to get frequent 1-itemset
 - Generate length candidate $(k+1)$ -itemsets from length k frequent itemsets
 - Test the candidates against DB
 - Terminate when no frequent candidates can be generated, else iterate

Types of Association Rules

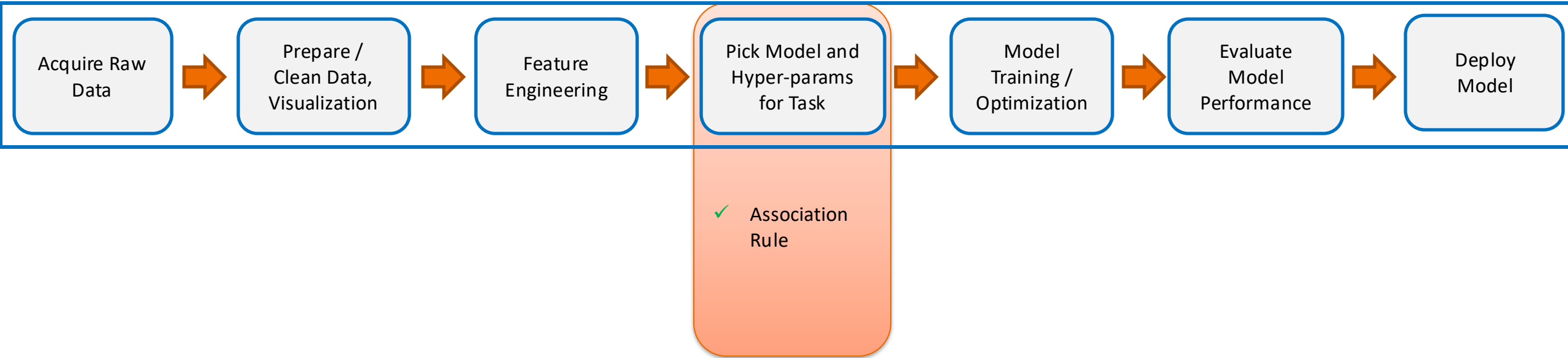
- Boolean association rules
- Hierarchical rules
 - dhoti, saree → t-shirt
- Quantitative & Categorical rules
 - (Age: 30...39), (Married: Yes) → (NumCars: 2)



More Types

- Cyclic / Periodic rules
 - Sunday → vegetables
 - Christmas → gift items
 - Summer, rich, jobless → ticket to Hawaii
- Constrained rules
 - Show itemsets whose average price > Rs.10,000
 - Show itemsets that have television on RHS
- Sequential rules
 - Star wars, Empire Strikes Back → Return of the Jedi

Summary



Thanks!!

Questions?